

Industry Information and the 52-Week High Effect^{*}

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ABSTRACT

We find that the 52-week high effect (George and Hwang, 2004) cannot be explained by risk factors. Instead, it is more consistent with investor underreaction caused by anchoring bias: the presumably more sophisticated institutional investors suffer less from this bias and buy (sell) stocks close to (far from) their 52-week highs. Further, the effect is mainly driven by investor underreaction to industry instead of firm-specific information. The extent of underreaction is more for positive than for negative industry information. A strategy that buys stocks in industries in which stock prices are close to 52-week highs and shorts stocks in industries in which stock prices are far from 52-week highs generates a monthly return of 0.60% from 1963 to 2009, roughly 50% higher than the profit from the individual 52-week high strategy in the same period. The 52-week high strategy works best among stocks with high R-squares and high industry betas (i.e., stocks whose values are more affected by industry factors and less affected by firm-specific information). Furthermore, our industry 52-week high effect is more pronounced among firms with less informative prices, exactly the type of firm on which investors are more likely to suffer from the anchoring bias. Our results hold even after controlling for both individual and industry return momentum effects.

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1. Introduction

The “52-week high effect” was first documented by George and Hwang (2004), who find that stocks with prices close to the 52-week highs have better subsequent returns than stocks with prices far from the 52-week highs. George and Hwang (2004) argue that investors use the 52-week high as an “anchor” against which they value stocks. When stock prices are near the 52-week high, investors are unwilling to bid the price all the way to the fundamental value. As a result, investors underreact when stock prices approach the 52-week high, and this creates the 52-week high effect.

In this paper, we show that an industry 52-week high trading strategy is more profitable than the individual 52-week high trading strategy proposed by George and Hwang (2004). Using all stocks listed on NYSE, AMEX, and NASDAQ from 1963 to 2009, a strategy that buys stocks in industries in which stock prices are close to 52-week highs and shorts stocks in industries in which stock prices are far from 52-week highs generates a monthly return of 0.60%, roughly 50% higher than the profit from the individual 52-week high strategy in the same period.

While the anchoring bias could be the reason behind the 52-week high effect, an alternative explanation is that stocks with prices close to 52-week highs are more risky than other stocks. To illustrate why risk factors can potentially cause the 52-week high effect, suppose that the market beta is the only risk factor. If the market return is high, high-beta stocks will have higher returns than other stocks and their prices are close to the 52-week highs. These stocks tend to have higher subsequent returns because market returns are positively correlated over time (see, e.g., Lo and MacKinlay, 1990). Conversely, if the market return is low, high-beta stocks will have lower returns and their prices are far from the 52-week highs. These stocks tend to

have lower subsequent returns because market returns are positively correlated over time. Therefore, we observe that stocks with prices close to their 52-week highs have higher subsequent returns than stocks with prices far from their 52-week highs, i.e., a 52-week high effect.

If the 52-week high effect is indeed caused by anchoring bias, then we would expect more sophisticated investors to suffer less from this bias and buy (sell) stocks whose prices are close to (far from) the 52-week highs. In contrast, less sophisticated investors should suffer more from this bias and trade in the opposite direction. On the other hand, if the 52-week high effect is driven by risk factors, then the trading strategy is no longer profitable after we properly control for different risks. Further, sophisticated investors should not buy (sell) stocks whose prices are close to (far from) the 52-week highs because the higher return is simply the compensation for higher risks associated with the trading strategy and there is no risk-adjusted abnormal return.

Many previous studies find that institutional investors are more sophisticated than individual investors (Gompers and Metrick, 2001; Cohen, Gompers, and Vuolteenaho, 2002; Sias, Starks, and Titman, 2006; Amihud and Li, 2006). Further, some studies find that institutional investors with short-term investment horizons trade actively to exploit the inefficiency in stock prices and they are more sophisticated than other institutional investors (Ke and Petroni, 2004; Lev and Nissim, 2006; Yan and Zhang, 2009). Therefore, we use institutional investors (especially transient ones) to proxy for sophisticated investors. We find that institutional investors buy (sell) stocks whose prices are close to (far from) the 52-week highs. The above pattern is more pronounced for transient institutional investors than for non-transient institutional investors.

We use either the Carhart (1997) four-factor model or the stock's mean return to control for potential risks associated with the 52-week high strategy, and find that the 52-week high effect still exists after the controls. The above evidence is more consistent with the underreaction explanation instead of the risk-based explanation.

We then go one step further in trying to understand what type of information that investors underreact to. Is the 52-week high effect driven mainly by investor underreaction to industry or firm-specific information? To positive or negative information? How can one design a better investment strategy based on the answers to the aforementioned questions? What are the implications of these findings on the efficient market hypothesis?

We find that the 52-week high effect is mainly driven by investor underreaction to industry instead of the firm-specific information. The individual 52-week high strategy used by George and Hwang (2004) works best among stocks with high R-squares and high industry betas (i.e., stocks whose values are more affected by industry factors and less affected by firm-specific information) and does not work among stocks with low R-squares and low industry betas. This suggests that the 52-week high effect is caused by industry instead of firm-specific information.

We also find that investor underreaction to positive news accounts more for the profits associated with the 52-week high strategy than investor underreaction to negative news. Given that it is positive news that pushes stock prices to their 52-week highs, the finding is not surprising. The Daniel, Grinblatt, Titman, and Wermers (1997; DGTW hereafter) benchmark-adjusted return for stocks in industries in which stock prices are close to 52-week highs is 0.28% per month, higher in magnitude than the -0.10% per month for stocks in industries in which stock prices are far from 52-week highs. This implies that the industry 52-week high strategy is highly

affected by costs associated with short-selling: the buy-only portfolio accounts for most of the profits. Our finding also casts doubt on the strong-form market efficiency hypothesis. Given that the trading strategy is based on publicly available information, and does not require extensive short-selling, why does the price not adjust to the information and eliminate the trading profits?

Our results may also offer insights on how to design better investment strategies based on 52-week highs. First, our results indicate that the individual 52-week high strategy proposed by George and Hwang (2004) is more profitable if one focuses on stocks with high industry betas and high R-squares. Second, investors can earn higher profits and bear lower risks if one buys (shorts) all stocks in industries whose stocks are close to (far from) 52-week highs instead of trading on individual price levels relative to their 52-week highs.

To provide more evidence that our industry 52-week high strategy is consistent with investor underreaction to public information due to anchoring bias, we divide firms into different groups based on how informative the firm's stock price is. We would expect investors to suffer more anchoring bias when the firm is hard to value and the stock price is less informative. We use five measures of price informativeness widely recognized in the literature: firm size, firm age, price impact, analyst coverage, and institutional ownership. Our industry 52-week high effect is more pronounced among firms whose stock prices are hard to value, namely, small firms, young firms, firms with large price impacts, firms with no analyst coverage, and firms with relatively low institutional ownership.

Following the prior literature (e.g., George and Hwang, 2004; Jegadeesh and Titman, 1993; Moskowitz and Grinblatt, 1999), we form equal-weighted portfolios when designing our industry 52-week high strategy. One criticism is that since we hold our portfolios for six months,

we need to rebalance our portfolios at the end of each month in order to keep it equal-weighted. The rebalancing can be potentially costly if the transaction cost is high. We address this issue by considering two variations in our strategy. First, we consider a modified industry 52-week high strategy in which we form an equal-weighted portfolio at the end of each month t , but do not rebalance in the next six months: i.e., we calculate the buy-and-hold return of the portfolio. Second, since we have shown that the industry 52-week high strategy is more profitable among small firms, investors can always implement the industry 52-week high strategy using only small stocks and form value-weighted portfolios. This way, investors do not have to worry about portfolio rebalancing, either. We find that the industry 52-week high strategy is still highly profitable using either of the above two modifications so that portfolio rebalancing is not necessary.

The rest of the paper is structured as follows. In section 2 we discuss related literature. In section 3 we describe data and sample selection. Section 4 presents all empirical results. Section 5 reports some robustness tests, and Section 6 concludes.

2. Related literature

Several recent studies have documented that the 52-week high has predictive ability for stock returns. George and Hwang (2004) find that the average monthly return for the 52-week high strategy is 0.45% from 1963 to 2001 and the return does not reverse in the long run. Li and Yu (forthcoming) examine the 52-week high effect on the aggregate market return. They use the nearness to the 52-week high and the nearness to the historical high as proxies for the degree of good news that traders have underreacted and overreacted in the past. For the aggregate market

returns, they find the nearness to the 52-week high positively predicts future market return, while the nearness to the historical high negatively predicts future returns. They also find that the predictive power from these proxies is stronger than traditional macro variables.

The 52-week high can not only predict future stock returns, it also affects mergers and acquisitions, exercise of options, mutual fund returns and flows, a stock's beta and return volatility, and trading volume. Baker, Pan, and Wurgler (2009) examine the 52-week high effect on mergers and acquisitions. They find that mergers and acquisitions offer prices that are biased toward the 52-week high, a highly salient but largely irrelevant past price, and the modal offer price is exactly that reference price. They also find that an offer's probability of acceptance discontinuously increases when the offer exceeds that 52-week high; conversely, bidder shareholders react increasingly negatively as the offer price is pulled upward toward that price.

The 52-week high price is not only the reference point for mergers and acquisitions, but also the reference point for the exercise of options. Heath, Huddart, and Lang (1999) investigate stock option exercise decisions by more than 50,000 employees at seven corporations. They find that employee exercise activity roughly doubles when the stock price exceeds the maximum price attained during the previous year. They interpret this as evidence that individual optionholders set a reference point based on the maximum stock price that was achieved within the previous year, and that they are more likely to exercise when subsequent price movements move them past their reference point.

Sapp (2011) documents the 52-week high effect on mutual fund returns and cash flows. He examines the performance of trading strategies for mutual funds based on an analogous 1-year high measure for the net asset value of fund shares, prior extreme returns, and fund

sensitivity to stock return momentum. He finds all three measures have significant, independent predictive power for fund returns, whether measured in raw or risk-adjusted returns. He also finds that nearness to the 1-year high is a significant predictor of fund monthly cash flows.

Driessen, Lin, and Hemert (2010) examine a stock's beta, return volatility, and option-implied volatility change when stock prices approach their 52-week highs and when stock prices break through these highs. They find that betas and volatilities decrease when stock prices approach 52-week highs, and that volatilities increase after breakthroughs. The effects are economically large and very significant, and consistent across stock and stock-option markets.

Huddart, Lang, Yetman (2008) examine the volume and price patterns around a stock's 52-week highs and lows. Based on a random sample of 2,000 firms drawn from the Center for Research in Security Price (CRSP) in the period from November 1, 1982, to December 31, 2006, they find that the volume is strikingly higher, in both economic and statistical terms, when the stock price crosses either the 52-week high or low. And this increase in volume is more pronounced the longer the time since the stock price last achieved the price extreme, the smaller the firm, and the higher the individual investor interest in the stock.

The 52-week high stock price is one of the most readily available aspects of past stock price behavior. For example, investors can find 52-week high stock price in *Yahoo Finance*, *Bloomberg*, and the *Wall Street Journal*. Why does such a simple and readily available measure affect financial market in so many ways? George and Hwang (2004) document that these effects are driven by investors' anchoring bias, which is based on the 52-week high price. Tversky and Kahneman (1974) discuss the concept of *anchoring*, which describe the common human tendency to rely too heavily on one piece of information when making decisions. George and

Hwang (2004) argue that investors use the 52-week high as an anchor when they evaluate new information: when the good news has pushed a stock's price near or to a new 52-week high, traders are reluctant to bid the price of the stock higher even if information warrants it. They argue that the information eventually prevails and the stock price moves up, resulting in a continuation.

Burghof and Prothmann (2009) test anchoring bias hypothesis. Motivated by a psychological insight, which states that behavioral biases increase under uncertainty, they examine whether the 52-week high price has more predictive power in cases of larger information uncertainty. Using firm size (market value), book-to-market ratio, the nearness to the 52-week high price, stock price volatility, firm age, and cash flow volatility as proxies for information uncertainty, they find that 52-week high strategy profits are increasing in uncertainty measures, which means that the anchoring bias hypothesis cannot be rejected.

3. Data and methodology

We design an industry 52-week high strategy based on the individual 52-week high strategy proposed by George and Hwang (2004). We first define *PRILAG* as

$$PRILAG_{i,t} = \frac{Price_{i,t}}{52weekhigh_{i,t}} \quad (1)$$

where $Price_{i,t}$ is the stock i 's price at the end of month t and $52weekhigh_{i,t}$ is the highest price of stock i during the 12-month period that ends on the last day of month t . The price information is

obtained from CRSP, and we use the corrections suggested in Shumway (1997).¹ The individual 52-week high strategy involves forming a portfolio at the end of each month t based on the value of $PRILAG_{i,t}$. The winner (loser) portfolio consists of 30% of stocks with the highest (lowest) value of $PRILAG_{i,t}$.

To construct the industry 52-week high strategy, we first use two-digit SIC codes to form 20 industries following Moskowitz and Grinblatt (1999).² In each month t , we calculate the weighted average of $PRILAG_{i,t}$ of all firms in an industry, where the weight is the market capitalization of the stock at the end of month t . The winners (losers) are stocks in the six industries with the highest (lowest) weighted averages of $PRILAG_{i,t}$.

In both individual and industry 52-week high strategies, we buy stocks in the winner portfolio and short stocks in the loser portfolio and hold them for six months. The return on the winner (loser) portfolio in month $t+k$ is the equal weighted return of all stocks in the portfolio, where $k=1, \dots, 6$. We compare the average monthly returns from July 1963 to December 2009 for these two strategies.

[Insert Table 1 here]

Results in Table 1 show that the individual 52-week high strategy generates an average monthly return of 0.43% in our sample period, close to the 0.45% documented in George and Hwang (2004) from July 1963 through December 2001. In contrast, the industry 52-week high strategy generates a monthly return of 0.60% (almost a 50% increase in profit compared to the

¹ Specifically, if a stock is delisted for performance reasons and the delist return is missing in CRSP, we set the delist return to -0.30 for NYSE/AMEX stocks and -0.55 for NASDAQ stocks. We obtain very similar results when we use only CRSP delist returns without filling missing performance related delist returns.

² See Table I in Moskowitz and Grinblatt (1999) for the description of the 20 industries.

individual 52-week high strategy), and the profit is statistically different from zero at the 1% level.

The returns to the individual and industry 52-week high strategies may be driven by certain firm characteristics. In particular, firms with prices close to their 52-week highs most likely have experienced high returns in the past several months and the profits could be due to the return momentum effect. To test whether this is the case, we use the DGTW benchmark-adjusted returns instead of raw returns. Specifically, we group stocks into 125 portfolios (quintiles based on size, book-to-market, and return momentum), and calculate the DGTW benchmark-adjusted return for a stock as its raw return minus the value-weighted average return of the portfolio to which it belongs.

The last three columns in Table 1 show that size, book-to-market ratio, and return momentum can indeed explain part of the profits generated by the two strategies. The average monthly profit of the individual 52-week high strategy is reduced to 0.08%, and not statistically different from 0. In contrast, we still have a sizeable 0.38% average monthly abnormal return associated with the industry 52-week high strategy, which remains highly statistically different from 0 (with a t -statistic of 5.22).

Most of the profits from the industry 52-week high strategy come from the buy portfolio. If one buys stocks in the six industries with the highest weighted averages of $PRILAG_{i,t}$, the average monthly DGTW benchmark-adjusted return is 0.28%. In contrast, the profit from shorting stocks in the six industries with the lowest weighted averages of $PRILAG_{i,t}$ is only 0.10%. Therefore, close to three fourths of the profits from the industry 52-week high strategy is generated by the buy portfolio. This has two implications. First, the industry 52-week high

strategy is highly implementable because most profits do not require shorting, which can be costly to implement. Second, it has implications on the efficient market hypothesis. Because 52-week highs are public information, why wouldn't investors simply buy more stocks in industries in which most stocks are close to their 52-week highs and drive away the abnormal returns?

George and Hwang (2004) document that the return to the 52-week high strategy is actually negative in January because loser stocks tend to rebound in January. Jegadeesh and Titman (1993) also document a negative return to the individual momentum strategy in January for the same reason. To examine whether the industry 52-week high strategy loses money in January, we exclude returns in January and repeat our analyses. Panel B shows that after excluding January, the profit to the individual 52-week high strategy increases dramatically, whereas the profit to the industry 52-week high strategy increases only slightly, especially for the DGTW benchmark-adjusted return. The results imply that the return to the individual 52-week high strategy is highly negative in January, whereas the profit to the industry 52-week high strategy is near 0 in January. The pattern is clearly borne out in Panel C, where we report the returns in January only. The profit to the individual 52-week high strategy is -7.62% (-1.90% based on DGTW benchmark-adjusted return) in January. The profit to the industry 52-week high strategy is -0.94% in January and insignificantly different from 0. The profit becomes positive (though not significantly different from 0) based on DGTW benchmark-adjusted return.

To summarize, we find that the industry 52-week high strategy is more profitable and less risky than the individual 52-week high strategy. The profit seems to be higher in the buy portfolio than in the short portfolio. Further, while the individual 52-week high strategy loses money in January, the industry 52-week high strategy does not.

4. Results

4.1. Can risk factors explain the industry 52-week high strategy?

While results in Tables 1 indicate that both individual and industry 52-week high strategies are profitable after controlling for size, book-to-market ratio, and momentum effects, we perform a more refined test with the Fama and French (1993) and Carhart (1997) four-factor model. In particular, the DGTW benchmark-adjusted return is a crude way of controlling for size, book-to-market ratio, and momentum effects because it essentially assumes that all firms in each of the 125 groups based on the three characteristics have the same factor loadings on the three factors, which may or may not be the case.

Specifically, we test for monthly abnormal returns on these portfolios as follows:

$$R_{p,t} - R_{f,t} = \alpha_p + b_p RMRF_t + s_p SMB_t + h_p HML_t + m_p MOM_t + e_{p,t}. \quad (2)$$

The dependent variable, $R_{p,t} - R_{f,t}$, is the monthly excess portfolio return, and $RMRF_t$, SMB_t , HML_t , and MOM_t are the Fama-French and Carhart factor portfolio returns. The intercept captures the average monthly abnormal performance. The data for the factor portfolio returns are from Wharton Research Data Service (WRDS).

[Insert Table 2 here]

Panel A in Table 2 shows that the average monthly abnormal return (after controlling for the four risk factors related to the market, size, book-to-market ratio, and momentum) of the winner portfolio based on the individual 52-week high strategy is 0.21%, which is statistically different from 0 (with a p -value less than 0.001). That of the loser portfolio is 0.07%, which is

not statistically different from 0. The profit to the long-short portfolio is 0.14% per month, not statistically different from 0.

Panel B in Table 2 shows that the average monthly abnormal return of the industry 52-week high strategy is 0.22%, which is statistically different from 0 (with a p -value of 0.029). Further, the profit from the industry 52-week high strategy comes entirely from the buy-only portfolio. The average monthly abnormal return of buying winners is 0.25%, which is highly statistically significant (with a p -value of 0.002). In contrast, the average monthly abnormal return of shorting losers is only 0.02%, which is not statistically different from 0.

[Insert Table 3 here]

There are potentially other risk factors that we do not capture in the four-factor model, and they could be related to the 52-week high strategy. To alleviate this concern, we use the mean monthly return of the stock in the sample period as the expected return on the stock. We define the mean-adjusted abnormal return on stock i in month t as the raw return minus the mean return on the stock. Panel A of Table 3 shows that the individual 52-week high strategy is not profitable any more, whereas the industry 52-week high strategy generates a monthly mean-adjusted abnormal return of 0.50%. In Panel B of Table 3, we exclude January returns, and find that both the individual and industry 52-week high strategies are profitable. As mentioned before, the negative profit in January is likely caused by tax effects. Panel C reports profits in January only. The individual and industry 52-week high strategies are losing 8.08% and 1.05% per month in Januarys, respectively.

To summarize, we use Fama and French (1993) and Carhart (1997) four-factor model and the firm's average return in the sample period to proxy for potential risk factors. We find that

these risk factors cannot explain the returns to individual or industry 52-week high effects. If our proxies capture all potential risk factors, then the evidence suggests that the 52-week high effect is unlikely to be caused by higher risks associated with the individual or industry 52-week high trading strategies.

4.2. Institutional demand and the 52-week high strategy

To further test whether the 52-week high effect is driven by anchoring bias or risk factors, we examine institutional demand according to a stock's closeness to the 52-week high. By definition, shares not held by institutional investors (more sophisticated) are held by individual investors (less sophisticated). While the anchoring bias hypothesis predicts that institutional investors buy (sell) stocks whose prices are close to (far from) 52-week highs, the risk factor hypothesis predicts no difference in institutional demand between the two groups of stocks.

We use two measures of institutional demand: the change in the fraction of shares held by institutional investors and the change in the number of institutions holding the stock. Because 13F reports institutional holdings each calendar quarter, we look at institutional demand change from quarter to quarter. At the end of quarter t , we rank stocks based on their closeness to the 52-week high (i.e., based on the value of *PRILAG*), and examine the average value of institutional demand change for firms in each group.

[Insert Table 4 here]

Panel A of Table 4 shows that, from quarter t to $t+1$, institutional investors increase their holdings of stocks whose prices are close to 52-week highs by 0.47% of the firm's shares outstanding. In contrast, they decrease their holdings of stocks whose prices are far from 52-week highs by 0.34%. The difference between the winner and loser groups is 0.81% and highly

statistically significant (with a t -statistic of 11.93). In the second subsequent quarter (from quarter $t+1$ to $t+2$), we find a similar pattern, though the magnitude is smaller, with a 0.56% difference between the winner and loser groups. The magnitude becomes even smaller in the third and fourth quarters, but there are still significant differences between the winner and loser groups.

The change in the number of institutions holding the firm's stocks shows a similar pattern. In quarter $t+1$, the number of institutional investors increases by 2.08 for stocks whose prices are close to 52-week highs. In contrast, the number decreases by 0.63 for stocks whose prices are far from 52-week highs. The difference between the winner and loser groups is highly statistically significant. In the next three quarters, we find a similar pattern, though the magnitude becomes smaller and smaller.

We then look at the trading by transient and non-transient institutional investors, separately. We use the definition of transient investors in Bushee (1998, 2001). Institutional investors in CDA/Spectrum are classified into transient, quasi-indexing, and dedicated investors based on their portfolio concentration and turnover rates. Transient institutions have high portfolio turnover and a diversified portfolio (see Bushee (1998, 2001) for details).³ We then calculate the fraction of a firm's shares held by transient investors and the number of transient institutions holding the firm's stock in each quarter.

Panel B of Table 4 shows that, in quarter $t+1$, transient institutional trading is 0.26% in the winner group and -0.19% in the loser group, and the difference is also highly statistically significant (with a t -statistic of 12.61). However, in quarter $t+2$, the difference in transient

³ We thank Brian Bushee for providing us with the dataset that classifies institutional investors into transient, quasi-indexing, and dedicated investors. We define non-transient institutions as quasi-indexing and dedicated investors.

institutional trading is much smaller between the top and bottom terciles. The pattern reverses in quarters $t+3$ and $t+4$. This may be due to the short investment horizon of transient institutions.

Panel C of Table 4 shows that, in quarter $t+1$, non-transient institutional trading is 0.19% in the winner group and -0.16% in the loser group, and the difference is statistically significant. The number of non-transient institutions holding the stock increases by 1.16 in the winner group and decreases by 0.41 in the loser group. The same pattern can be found in the next three quarters for both non-transient institutional trading and the number of non-transient institutions holding the stock.

[Insert Table 5 here]

In Table 5, we repeat our analyses in Table 4 but rank stocks at the end of each quarter based on industry closeness to the 52-week high, and define the winner (loser) group as the six industries with the highest (lowest) values of the industry weighted average of *PRILAG*.

Panel A of Table 5 shows that, in quarter $t+1$, institutional investors increase their holdings of stocks whose prices are close to 52-week highs by 0.24% of the firm's shares outstanding. In contrast, they increase their holdings of stocks whose prices are far from 52-week highs by only 0.14%. The difference between the winner and loser groups is 0.10% and statistically significant at the 5% level. In the next three quarters, institutional investors increase their holdings more in winner industries than in loser industries, but the difference is not statistically significant.

The change in the number of institutions holding the firm's stocks is more for stocks in winner industries than for stocks in loser industries. In quarter $t+1$, the number of institutional investors increases by 1.27 for stocks in winner industries. In contrast, the number increases by

only 0.35 for stocks in loser industries. The difference between the winner and loser groups is statistically significant. In the next three quarters, we find a similar pattern, though the magnitude becomes smaller.

Panel B of Table 5 shows that, in the first subsequent quarter, transient institutions increase their holding of stocks in winner industries by 0.08% and stocks in loser industries by 0.00%, and the difference between the two groups is statistically different from 0 at the 5% level. Institutional trading is not higher for stocks in winner industries than those in loser industries in quarters $t+2$ through $t+4$. This may be due to the short investment horizon of transient institutions. The change in the number of institutions holding the firm's stocks shows a similar picture. The difference in the change in the number of transient institutions between the winner and loser industries is statistically significant in quarter $t+1$, but not in the next three quarters.

Panel C of Table 5 shows that, in quarters $t+1$ and $t+2$, there is no difference in non-transient institutional trading between stocks in winner and loser industries. In quarters $t+3$ and $t+4$, however, we find that non-transient institutions increase their holdings of stocks in winner industries more than stocks in loser industries. The number of non-transient institutions holding the stock increases more for stocks in winner industries than for stocks in loser industries in quarters $t+1$ to $t+4$.

To summarize, we find that institutional investors, especially transient ones, generally increases their holding of stocks whose prices are close to 52-week highs and decreases their holding of stocks whose prices are far from 52-week highs. This seems to support the anchoring bias hypothesis instead of the risk-based explanation for the 52-week high effect.

4.3. Can return momentum explain the industry 52-week high strategy?

Because there is a positive correlation between past returns and closeness to the 52-week high, and Moskowitz and Grinblatt (1999) show that return momentum is mainly driven by industry information, one may wonder whether the profit from the industry 52-week high strategy is caused by the momentum in industry information. To test this, we construct two momentum strategies: the individual momentum strategy proposed by Jegadeesh and Titman (1993) and the industry momentum strategy proposed by Moskowitz and Grinblatt (1999).

The winners (losers) in the individual momentum strategy are the 30% of stocks with the highest (lowest) returns in the past six months. To construct the industry momentum strategy, we calculate industry return as the value-weighted return of all firms in the industry every month for each of the 20 industries. The winners (losers) are stocks in the six industries with the highest (lowest) cumulative industry returns in the past six months. In both individual and industry momentum strategies, we buy stocks in the winner portfolio and short stocks in the loser portfolio and hold them for six months. The return on the winner (loser) portfolio in month t is the equal weighted return of all stocks in the portfolio.

[Insert Table 6 here]

We first perform a pairwise comparison between individual momentum strategy and the industry 52-week high strategy. In Panel A of Table 6, we first group firms into winners, losers, and the middle group (the rest) based on individual momentum strategy. Then within each group, we perform the industry 52-week high strategy by buying (shorting) stocks in the six industries with the highest (lowest) industry average of $PRILAG_{i,t}$. We can see that the industry 52-week high strategy is profitable in each group. In contrast, when we first group firms into winners, losers, and the middle group based on the industry 52-week high strategy in Panel B, the

individual momentum strategy is not always profitable: the strategy is not profitable in the winner or middle group based on DGTW benchmark-adjusted returns.

In Panels C and D, we do a pairwise comparison between industry momentum strategy and the industry 52-week high strategy. If we group firms into winners, losers, and the middle group based on industry momentum strategy, the industry 52-week high strategy is profitable in each group (with the only exception of the loser group when we use raw returns). When we group firms into winners, losers, and the middle group based on the industry 52-week high strategy, the industry momentum strategy is also profitable in each group.

Results in Panels A through D show that the industry 52-week high strategy is not subsumed by either the individual or the industry return momentum effect. We also perform a pairwise comparison between individual and industry 52-week high strategies. Panels E and F report results. If we group firms into winners, losers, and the middle group based on individual 52-week high strategy, the industry 52-week high strategy is profitable in each group. When we group firms into winners, losers, and the middle group based on the industry 52-week high strategy, the individual 52-week high strategy is profitable only in the loser group.

4.4. Compare the four strategies simultaneously

Following Fama-MacBeth (1973) and George and Hwang (2004), we run the following regression to compare the four strategies simultaneously and control for the effects of firm size and bid-ask bounce:

$$\begin{aligned}
 R_{i,t} = & b_{0jt} + b_{1jt} R_{i,t-1} + b_{2jt} SIZE_{i,t-1} + b_{3jt} JH_{i,t-j} + b_{4jt} JL_{i,t-j} + b_{5jt} MH_{i,t-j} + b_{6jt} ML_{i,t-j} + b_{7jt} GH_{i,t-j} \\
 & + b_{8jt} GL_{i,t-j} + b_{9jt} IH_{i,t-j} + b_{10jt} IL_{i,t-j} + e_{p,t}.
 \end{aligned} \tag{3}$$

The dependent variable, $R_{i,t}$, is the return to stock i in month t . We skip one month between portfolio forming month and holding period and include month $t-1$ return $R_{i,t-1}$ in the regression to control for the effect of bid-ask bounce. Because we form portfolio every month and hold the portfolio for six months, the profit from a winner or loser portfolio in month t can be calculated as the sum of returns to six portfolios, each formed in one of the six past successive months $t-j$, where $j=2, 3, \dots, 7$ (we skip one month between portfolio formation and holding). $JH_{i,t-j}$ is a dummy variable with value 1 if stock i is included in the Jegadeesh and Titman (1993) winner portfolio in month $t-j$ (i.e., if the stock is in the top 30% based on returns from month $t-j-6$ to month $t-j$); and 0 otherwise. Similarly, $JL_{i,t-j}$ is a dummy variable indicating whether stock i is included in the Jegadeesh and Titman (1993) loser portfolio in month $t-j$. $MH_{i,t-j}$ and $ML_{i,t-j}$ are dummy variables for Moskowitz and Grinblatt (1999) industry momentum winner and loser portfolios, and $GH_{i,t-j}$ and $GL_{i,t-j}$ are dummy variables for George and Hwang (2004) individual 52-week high winner and loser portfolios. For our industry 52-week high winner and loser portfolios, we create two dummies, $IH_{i,t-j}$ and $IL_{i,t-j}$.

Following George and Hwang (2004), we first run separate cross-sectional regressions of equation (3) for each $j=2, \dots, 7$. Then the total return in month t of a portfolio is the average over $j=2, \dots, 7$. For example, the month t return to the Jegadeesh and Titman (1993) individual momentum winner portfolio is $\frac{1}{6} \sum_{j=2}^7 b_{3jt}$. We then report in Table 4 the time-series averages of these values and the associated t -statistics when either the raw return or the DGTW benchmark-adjusted return is the dependent variable. Profits from the four investment strategies are reported in the bottom panel. We also run regressions excluding Januarys and in Januarys only.

[Insert Table 7 here]

When we use raw return as the dependent variable, the industry 52-week high strategy generates a return of 0.34% after controlling for the other three investing strategies, indicating that the profits from the industry 52-week high are above and beyond those from the other three strategies. Results excluding Januarys are similar. The third column shows that, in Januarys, only the industry 52-week high strategy generates profits, and none of the other three does. The results using DGTW benchmark-adjusted returns are similar. In particular, the industry 52-week high strategy generates an abnormal return of 0.26% after controlling for the profits from other three investing strategies; further, the magnitude of the profits from the industry 52-week high is greater than that from any of the other three strategies.

4.5. Do profits from the industry 52-week high strategy reverse in the long run?

To test whether the industry 52-week high strategy is consistent with investor underreaction to industry information, we examine the long-run returns to this strategy. We run a regression similar to that in equation (3), but we skip more than one month between portfolio formation and the holding period. To analyze the return 12 months after portfolio formation, we define $JH_{i,t-j}$ as a dummy with value 1 if stock i is included in the Jegadeesh and Titman (1993) winner portfolio in month $t-j$, where $j=13, \dots, 18$; and 0 otherwise. Other dummies are defined similarly using $j=13, \dots, 18$. To analyze the return 24 and 36 months after portfolio formation, we change the value of j to $j=25, \dots, 30$ and $j=37, \dots, 42$.

[Insert Table 8 here]

Results in Table 8 show that there is no return reversal for the industry 52-week high strategy. However, there is evidence of long-run return reversal for the individual momentum strategy. There is also some weak evidence of long-run return reversal for the individual 52-

week high strategy when we use the DGTW benchmark-adjusted returns (though there is no reversal if we use raw returns, as reported in George and Hwang (2004)).

4.6. *Is the 52-week high effect driven by industry or firm-specific information?*

So far, our results show that industry 52-week high strategy is more profitable and less risky than individual 52-week high strategy. This suggests that the 52-week high effect is mainly driven by investor underreaction to industry information. If this is true, then the 52-week high effect documented by George and Hwang (2004) should be more pronounced among firms whose values are influenced more by industry information and less by firm-specific information, i.e., stocks with high industry betas and high R-squares.

To estimate industry beta and R-square, we run the following regression for each stock i using daily stock return data in the past 12 months:

$$R_{i,t} = \alpha_i + \beta_{mkt,i} R_{m,t} + \beta_{ind,i} R_{ind,t} + e_{i,t}, \quad (4)$$

where $R_{m,t}$ is the market return at day t , and $R_{ind,t}$ is the value-weighted return of all stocks in stock i 's industry at day t . To avoid any spurious results, the industry portfolio is constructed without stock i . Industry beta is the estimated value of $\beta_{ind,i}$, and R-square is the adjusted R-square from the above regression. At the end of each month, we repeat the above regression and rank stocks based on industry beta and R-square. We then examine the profits to the individual 52-week high strategy in each industry beta tercile and R-square tercile.

[Insert Table 9 here]

Panel A of Table 9 shows that the profit to the individual 52-week high strategy is 0.32% per month among firms with the lowest industry betas. The profit increases to 0.40% in the

middle group and 0.51% among firms with the highest industry betas. Results based on DGTW benchmark-adjusted returns show a similar pattern. The 52-week high effect is the strongest among high industry beta firms and the weakest among low industry beta firms.

Panel B of Table 9 shows that the profit to the individual 52-week high strategy increases with a firm's R-square. The profit among firms in the lowest tercile of R-square is -0.05% per month, though not statistically significant. The profit increases to 0.56% in the middle group and 0.80% among firms with the highest R-squares. If we use DGTW benchmark-adjusted returns, the individual 52-week high strategy actually loses 0.26% per month among firms with the lowest R-squares, and the negative profit is statistically different from 0 at the 5% level. The profit increases to 0.16% in the middle group and 0.33% among firms with the highest R-squares.

4.7. Price informativeness and the industry 52-week high effect

If the profits from the industry 52-week high strategy are indeed driven by the anchoring bias of investors, we should expect the bias to be stronger among firms whose stock prices are hard to value. Therefore, the industry 52-week high effect should be more (less) pronounced among firms with less (more) informative prices. To test this, we use five price informativeness measures that are widely recognized in the literature. The five measures are as follows.

1. Firm size, defined as the firm's market capitalization at the end of the month of the portfolio formation. It is well known that large firms have more informative prices than small firms.
 2. Firm age, measured as the number of months during which the stock is publicly traded.
- Availability of public trading history will surely reduce the information asymmetry

between the firm and outside investors. Therefore, older firms should have more informative prices than younger firms.

3. Price impact, measured by the absolute daily return divided by the daily dollar volume of trade (in millions), averaged over the past twelve months, similar to the definition in Amihud (2002). It measures how easily investors can liquidate a stock without severely affecting the price. Firms with less informative prices usually have high price impacts.
4. Analyst coverage, defined as the number of analysts following the firm. Firms with more analyst coverage have more informative prices.
5. Institutional ownership, defined as the fraction of shares held by institutions who file the 13F form with the Securities and Exchange Commission. It is widely accepted that institutional investors are more sophisticated than retail investors. Therefore, firms with more institutional ownership have less information asymmetry.

We divided firms into three groups based on each of the above measures, and perform the industry 52-week high strategy, and report the profit to the strategy in each group. Table 10 reports the results.

[Insert Table 10 here]

Panel A of Table 10 shows that the profit to the industry 52-week high strategy is 0.74% per month among small firms (the bottom 1/3 of firms based on firm size). In contrast, the profit is 0.66% among mid-sized firms and 0.41% among large firms. Results based on DGTW benchmark-adjusted returns show a similar pattern. If we look at the alpha from the four-factor models, the profit is 0.41% among small firms and statistically different from 0 at the 1% level. The profit is essentially 0 among large firms.

Panel B of Table 10 shows that the profit to the industry 52-week high strategy decreases with a firm's age. The profit among firms in the bottom tercile is 0.76% per month, 0.66% in the middle tercile and 0.40% in the top tercile. If we use DGTW benchmark-adjusted returns, the profit is 0.46% per month among young firms, and 0.29% for old firms. The four-factor models produce the same pattern, and the alpha is 0.28% for young firms and an insignificant 0.10% for old firms.

Panels C, D, and E, report results based on price impact, analyst coverage, and institutional ownership, respectively. They all show the same pattern. The industry 52-week high strategy is more profitable among firms with high information asymmetry (firms with high price impact, no analyst coverage, and low institutional ownership). The results in Table 10 are consistent with the notion that the industry 52-week high effect is driven by investors' anchoring bias.

4.8. Portfolio rebalancing and the industry 52-week high strategy

So far, we have followed the prior literature (e.g., George and Hwang, 2004; Jegadeesh and Titman, 1993; Moskowitz and Grinblatt, 1999) and formed equal-weighted portfolios when designing our strategies. One criticism is that since we hold our portfolios for six months, we need to rebalance our portfolios at the end of each month in order to keep it equal-weighted. The rebalancing can be potentially costly if the transaction cost is high, and it is not clear whether our strategies are still profitable after transaction costs. We address the implementability of the industry 52-week high strategy related to the rebalancing of the portfolio in this subsection.

First, we consider a modified industry 52-week high strategy that does not require monthly portfolio rebalancing. Specifically, at the end of each month t , we buy an equal-

weighted portfolio of stocks in the six industries with the highest values of industry *PRILAG*, and short the same dollar amount of an equal-weighted portfolio of stocks in the six industries with the lowest values of industry *PRILAG*. We then hold the portfolio for six months without rebalancing. Therefore, at the end of months $t+1$ through $t+5$, the portfolio is neither equal-weighted nor value-weighted. To calculate the average monthly return of such a strategy, we first calculate the six-month cumulative buy-and-hold raw return of each stock in each portfolio. The cumulative profit of the modified industry 52-week high strategy (call it CRET) is the mean cumulative return of all stocks in the long portfolio minus that of all stocks in the short portfolio. The monthly profit of the modified industry 52-week high strategy is then $(1+CRET)^{1/6}-1$.

To calculate the abnormal return of the above modified industry 52-week high strategy, we form 125 portfolios at the end of month t based on size, book-to-market ratio, and momentum. The six-month cumulative abnormal return of each stock is the cumulative raw return minus the cumulative return on one of the 125 portfolios to which the stock belongs. The cumulative abnormal return of the modified industry 52-week high strategy (call it ACRET) is the mean abnormal cumulative return of all stocks in the long portfolio minus that of all stocks in the short portfolio. The monthly abnormal return of the modified industry 52-week high strategy is then $(1+ACRET)^{1/6}-1$. The modified individual 52-week high strategy is similar, except that the winner and loser portfolios are formed based on individual *PRILAG* rather than industry *PRILAG*.

[Insert Table 11 here]

Panel A of Table 11 shows that the modified industry 52-week high strategy that does not require monthly rebalancing is still profitable, with an average monthly return of 0.66%, similar to the 0.60% we have documented in Panel A of Table 2. The average DGTW benchmark

adjusted abnormal return of the strategy is 0.39% per month, which is greater than the abnormal return of 0.09% on the modified individual 52-week high strategy.

We now consider the second way to address the rebalancing concern. In Table 10, we have seen that the industry 52-week high strategy is more profitable among small firms. If investors want to implement the industry 52-week high strategy, they can always focus on small stocks and form value-weighted portfolios. This way, investors do not have to worry about portfolio rebalancing. To see if such a strategy is still profitable, we buy a value-weighted portfolio of small stocks in the six industries with the highest values of industry *PRILAG*, and short the same dollar amount of a value-weighted portfolio of small stocks in the six industries with the lowest values of industry *PRILAG*. Small stocks are defined as the 25% of stocks with the lowest values of market capitalization at the end of month t . Similarly, we calculate the profit of the individual 52-week high strategy among small stocks using value-weighted portfolios.

Panel B of Table 11 shows that the industry 52-week high strategy is still profitable if we focus on small stocks and use value-weighted portfolios, with an average monthly return of 0.93%. The average DGTW benchmark adjusted abnormal return of the strategy is 0.52% per month, which is greater than the abnormal return of 0.33% on the corresponding individual 52-week high strategy.

To summarize, even though we follow the literature and form equal-weighted portfolios in our industry 52-week high strategy, which requires monthly rebalancing of the portfolio, our results still hold if we modify our strategy so that portfolio rebalancing is not necessary.

5. Robustness tests

In this section, we perform some robustness tests regarding our main findings.

5.1. Sample periods

To test if our results hold over different time periods, we divide our sample period into three sub-periods: July 1963 to December 1978, January 1979 to December 1994, and January 1995 to December 2009, so that each sub-period has roughly the same length. We compare the profits to the individual and industry 52-week high strategies in each sub-period, using both raw returns and DGTW benchmark-adjusted returns.

[Insert Table 12 here]

Table 12 shows that from July 1963 to December 1978, the individual 52-week high strategy generates 0.08% per month, which is insignificantly different from 0. In contrast, the industry 52-week high strategy generates 0.38% per month, a value highly significantly different from 0. When we use DGTW benchmark-adjusted returns, the industry 52-week high strategy still generates significant profits, whereas the profits to the individual 52-week high strategy are not statistically significant.

From January 1979 to December 1994, when we use raw returns, both the individual and industry 52-week high strategies generate significant profits, though the profit from the individual 52-week high strategy is slightly higher. However, when we use DGTW benchmark-adjusted returns, only the industry 52-week high strategy generates significant profits, whereas the individual 52-week high strategy does not produce statistically significant profits. From January 1995 to December 2009, the industry 52-week high strategy generates significant profits

based on either raw returns or DGTW benchmark-adjusted returns. In contrast, the individual 52-week high strategy generates no significant profits when we use either raw returns or the DGTW benchmark-adjusted returns.

The above results show that in each sub-period, the industry 52-week high strategy generates more profits than the individual 52-week high strategy. We also explore whether our results are driven by the extreme market conditions. Specifically, during the internet bubble period, most stocks have very high stock prices and prices are either at or close to their 52-week highs. In contrast, during the recent financial crisis, most stocks have very low prices that are far from their 52-week highs. We test if our results are robust to the exclusion of the following two periods: 1998-2000 and 2008-2009.

Results at the bottom of Table 12 show that our results hold even after excluding the internet bubble period and the recent financial crisis period. The individual 52-week high strategy generates 0.48% per month, which is significantly different from 0 at the 5% level. The industry 52-week high strategy generates 0.51% per month, which is significantly different from 0 at the 1% level. When we use DGTW benchmark-adjusted returns, the difference between the profits from the two strategies widens. The individual 52-week high strategy generates an insignificant 0.10% per month, whereas the industry 52-week high strategy generates 0.28% per month and it is statistically significant at the 1% level.

5.2. Changing the holding period to three or twelve months

We follow George and Hwang (2004) and hold the portfolios for six months after forming the winner and loser portfolios. We examine whether our results hold if we hold the portfolio for three or twelve months. Results are reported in Table 13.

[Insert Table 13 here]

Panel A of Table 13 shows that if we hold the portfolios for three months instead of six months, the individual 52-week high strategy generates 0.44% per month, whereas the industry 52-week high strategy generates 0.78% per month. When we use DGTW benchmark-adjusted returns, only the industry 52-week high strategy generates significant profits, whereas the individual 52-week high strategy does not produce statistically significant profits. By looking at profits excluding Januarys and in Januarys only, we can see that there is a large negative return for the individual 52-week high strategy in January, whereas the profits to the industry 52-week high is insignificantly different from 0 in January.

Results in Panel B show that if we hold the portfolios for twelve months, the industry 52-week high strategy generates more profits than the individual 52-week high strategy, measured by either raw returns or DGTW benchmark-adjusted returns. By looking at profits in January only, we can still see a large negative return for the individual 52-week high strategy. The profit to the industry 52-week high is also negative in January if we use raw returns. However, if we use DGTW benchmark-adjusted returns, there is no negative profit associated with the industry 52-week high.

Table 13 shows that if we hold our portfolios for three or twelve months instead of six months, industry 52-week high strategy is still more profitable than individual 52-week high strategy.

6. Conclusion

In this paper, we find that the 52-week high effect (George and Hwang, 2004) cannot be explained by risk factors, where we use either Fama and French (1993) and Carhart (1997) four-factor model or a stock's mean return in our sample period to proxy for risk factors. We find that it is more consistent with investor underreaction caused by anchoring bias: the presumably more sophisticated institutional investors suffer less from this bias and buy (sell) stocks close to (far from) their 52-week highs. Further, the 52-week high effect is mainly driven by investor underreaction to industry information. The extent of underreaction is more for positive than for negative industry information. A strategy that buys stocks in industries in which stock prices are close to 52-week highs and shorts stocks in industries in which stock prices are far from 52-week highs generates a monthly return of 0.60% from 1963 to 2009, roughly 50% higher than the profit from the individual 52-week high strategy in the same period. The 52-week high strategy works best among stocks with high R-squares and high industry betas (i.e., stocks whose values are most affected by industry factors and least affected by firm-specific information). Our results hold even after controlling for both individual and industry momentum effects.

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Table 1: Profits from individual and industry 52-week high strategies

This table reports the average monthly portfolio returns from July 1963 through December 2009 for individual and industry 52-week high strategies. All portfolios are held for 6 months. The winner (loser) portfolio in the individual 52-week high strategy is the equally weighted portfolio of the 30% stocks with the highest (lowest) ratio of current price to 52-week high. The winner (loser) portfolio in the industry 52-week high strategy is the equally weighted portfolio of stocks in the top (bottom) 6 industries ranked by the industry value-weighted ratio of current price to 52-week high. The sample includes all stocks in CRSP; *t*-statistics are in parentheses.

Panel A: All months included						
	Raw return			DGTW return		
	Winner	Loser	Winner-Loser	Winner	Loser	Winner-Loser
Individual	1.35%	0.92%	0.43%	0.11%	0.03%	0.08%
	(7.38)	(2.57)	(1.81)	(4.00)	(0.58)	(1.10)
Industry	1.47%	0.87%	0.60%	0.28%	-0.10%	0.38%
	(6.28)	(3.19)	(4.72)	(6.35)	(-2.36)	(5.22)
Panel B: Excluding January						
	Raw return			DGTW return		
	Winner	Loser	Winner-Loser	Winner	Loser	Winner-Loser
Individual	1.21%	0.05%	1.16%	0.13%	-0.10%	0.26%
	(6.42)	(0.15)	(5.66)	(6.11)	(-2.19)	(3.88)
Industry	1.11%	0.37%	0.74%	0.26%	-0.14%	0.41%
	(4.72)	(1.39)	(5.90)	(5.87)	(-3.16)	(5.33)
Panel C: January only						
	Raw return			DGTW return		
	Winner	Loser	Winner-Loser	Winner	Loser	Winner-Loser
Individual	2.95%	10.57%	-7.62%	-0.44%	1.45%	-1.90%
	(4.15)	(6.06)	(-5.73)	(-3.37)	(6.00)	(-5.55)
Industry	5.50%	6.45%	-0.94%	0.44%	0.33%	0.11%
	(5.89)	(5.11)	(-1.52)	(2.45)	(1.97)	(0.43)

Table 2: Regression results of monthly returns on individual and industry 52-week high stock portfolios

We run the following four-factor model:

$$R_{p,t} - R_{f,t} = a_p + b_p RMRF_t + s_p SMB_t + h_p HML_t + m_p MOM_t + e_{p,t}.$$

The dependent variable is the equal-weighted monthly portfolio excess return ($R_{p,t} - R_{f,t}$) on individual (Panel A) or industry (Panel B) 52-week high stock portfolios. The winner (loser) portfolio in individual 52-week high strategy is the equally weighted portfolio of the 30% stocks with the highest (lowest) ratio of current price to 52-week high. The winner (loser) portfolio in industry 52-week high strategy is the equally weighted portfolio of the stocks in the top (bottom) six industries ranked by the industry value-weighted ratio of current price to 52-week high. All portfolios are held for 6 months. $RMRF_t$ is the realized market risk premium; SMB_t is the excess return of a portfolio of small stocks over a portfolio of big stocks; HML_t is the excess return of a portfolio of high book-to-market ratio stocks over a portfolio of low book-to-market ratio stocks; and MOM_t is the excess return on the prior-period winner portfolio over the prior-period loser portfolio. The monthly realizations for the three Fama-French factors and for Carhart's momentum factor are from WRDS. Portfolio raw returns and the excess returns over the risk-free rate are shown under the headings *Raw ret.* and *Excess ret.* Regression coefficients are reported with *p*-values in italics. The monthly portfolio return data are from July 1963 to December 2009.

Panel A: Individual 52-week high portfolios							
	Raw ret.	Excess ret.	Intercept	RMRF	SMB	HML	MOM
Winner	0.0135	0.0089	<i>0.0021</i> <i><0.001</i>	<i>0.8194</i> <i><0.001</i>	<i>0.5093</i> <i><0.001</i>	<i>0.2433</i> <i><0.001</i>	<i>0.1629</i> <i><0.001</i>
Middle	0.0119	0.0074	<i>0.0012</i> <i>0.016</i>	<i>0.9304</i> <i><0.001</i>	<i>0.7446</i> <i><0.001</i>	<i>0.2633</i> <i><0.001</i>	<i>-0.0997</i> <i><0.001</i>
Loser	0.0092	0.0046	<i>0.0007</i> <i>0.66</i>	<i>1.0453</i> <i><0.001</i>	<i>1.3076</i> <i><0.001</i>	<i>0.0614</i> <i>0.247</i>	<i>-0.5509</i> <i><0.001</i>
Winner - Loser	0.0043		<i>0.0014</i> <i>0.352</i>	<i>-0.2260</i> <i><0.001</i>	<i>-0.7983</i> <i><0.001</i>	<i>0.1818</i> <i><0.001</i>	<i>0.7138</i> <i><0.001</i>
Panel B: Industry 52-week high portfolios							
	Raw ret.	Excess ret.	Intercept	RMRF	SMB	HML	MOM
Winner	0.0147	0.0102	<i>0.0025</i> <i>0.002</i>	<i>0.9415</i> <i><0.001</i>	<i>0.7884</i> <i><0.001</i>	<i>0.3005</i> <i><0.001</i>	<i>0.0721</i> <i><0.001</i>
Middle	0.0120	0.0075	<i>0.0014</i> <i>0.052</i>	<i>0.9382</i> <i><0.001</i>	<i>0.8446</i> <i><0.001</i>	<i>0.2427</i> <i><0.001</i>	<i>-0.1356</i> <i><0.001</i>
Loser	0.0087	0.0042	<i>0.0002</i> <i>0.804</i>	<i>0.9504</i> <i><0.001</i>	<i>0.8659</i> <i><0.001</i>	<i>0.0909</i> <i>0.008</i>	<i>-0.3677</i> <i><0.001</i>
Winner - Loser	0.0060		<i>0.0022</i> <i>0.029</i>	<i>-0.0089</i> <i>0.712</i>	<i>-0.0775</i> <i>0.017</i>	<i>0.2096</i> <i><0.001</i>	<i>0.4398</i> <i><0.001</i>

Table 3: Mean-adjusted returns for individual and industry 52-week high strategies

This table reports the average monthly portfolio mean-adjusted returns from July 1963 through December 2009 for individual and industry 52-week high strategies. The mean-adjusted return of stock i at month t is defined as the raw return of stock i in month t minus the average monthly return of stock i from 1963 to 2009. All portfolios are held for 6 months. The winner (loser) portfolio in the individual 52-week high strategy is the equally weighted portfolio of the 30% stocks with the highest (lowest) ratio of current price to 52-week high. The winner (loser) portfolio in industry 52-week high strategy is the equally weighted portfolio of the stocks in the top (bottom) 6 industries ranked by the industry value-weighted ratio of current price to 52-week high. The sample includes all stocks on CRSP; t -statistics are in parentheses.

Panel A: All months included			
	Winner	Loser	Winner-Loser
Individual	0.03%	0.08%	-0.05%
	(0.17)	(0.22)	(-0.20)
Industry	0.27%	-0.23%	0.50%
	(1.17)	(-0.84)	(3.94)
Panel B: Excluding January			
	Winner	Loser	Winner-Loser
Individual	-0.11%	-0.79%	0.67%
	(-0.60)	(-2.41)	(3.34)
Industry	-0.09%	-0.73%	0.64%
	(-0.38)	(-2.79)	(5.13)
Panel C: January only			
	Winner	Loser	Winner-Loser
Individual	1.63%	9.71%	-8.08%
	(2.32)	(5.55)	(-6.02)
Industry	4.30%	5.35%	-1.05%
	(4.63)	(4.23)	(-1.67)

Table 4: Institutional demand in individual 52-week high portfolios

This table reports quarterly changes in total, transient, and non-transient institutional holding and changes in the number of total, transient, and non-transient institutional investors holding the stocks in individual 52-week high portfolios. Total, transient, or non-transient institutional holding of a stock in a quarter is defined as the number of shares held by all, transient, or non-transient institutional investors at the end of that quarter divided by the number of shares outstanding. For each quarter t , we group all stocks into three individual 52-week high portfolios. The individual 52-week high winner (loser) portfolio is the equally weighted portfolio of the 30% stocks with the highest (lowest) ratio of current price to 52-week high. The individual 52-week high middle portfolio is the equally weighted portfolios of stocks that are neither individual 52-week high winners nor losers. For each portfolio, we report quarterly equal-weighted average of change in institutional holding and change in the number of institutions holding the stock for quarters $t+1$ to $t+4$. Panels B & C report the results for transient and non-transient institutions, respectively. The classification of institutions is from Brian Bushee's website. The t -statistics are in parentheses.

Panel A: All institution investors								
	Change in institutional holding				Change in investor number			
	Loser	Middle	Winner	W - L	Loser	Middle	Winner	W - L
$t + 1$	-0.34%	0.45%	0.47%	0.81%	-0.63	0.81	2.08	2.71
	(-3.93)	(4.49)	(5.07)	(11.93)	(-4.57)	(4.33)	(8.38)	(12.84)
$t + 2$	-0.18%	0.31%	0.39%	0.56%	-0.19	0.81	1.59	1.78
	(-2.04)	(3.07)	(4.08)	(8.32)	(-1.39)	(4.36)	(6.56)	(9.81)
$t + 3$	-0.07%	0.24%	0.30%	0.37%	0.00	0.79	1.37	1.37
	(-0.79)	(2.32)	(3.03)	(5.19)	(0.03)	(4.15)	(5.62)	(7.58)
$t + 4$	0.01%	0.21%	0.18%	0.17%	0.15	0.75	1.19	1.04
	(0.13)	(2.07)	(1.80)	(2.34)	(1.11)	(3.88)	(4.79)	(5.84)

Panel B: Transient institutional investors								
	Change in institutional holding				Change in investor number			
	Loser	Middle	Winner	W - L	Loser	Middle	Winner	W - L
$t + 1$	-0.19%	0.07%	0.26%	0.46%	-0.25	0.16	0.81	1.06
	(-4.87)	(1.62)	(6.67)	(12.61)	(-3.58)	(1.78)	(7.28)	(11.39)
$t + 2$	-0.01%	0.06%	0.08%	0.10%	0.04	0.26	0.39	0.35
	(-0.34)	(1.46)	(1.95)	(2.70)	(0.61)	(2.84)	(3.73)	(5.06)
$t + 3$	0.07%	0.05%	-0.00%	-0.07%	0.14	0.28	0.27	0.13
	(1.62)	(1.36)	(-0.10)	(-1.87)	(1.81)	(3.04)	(2.51)	(1.90)
$t + 4$	0.11%	0.05%	-0.06%	-0.17%	0.21	0.29	0.17	-0.04
	(2.87)	(1.33)	(-1.39)	(-5.02)	(2.87)	(3.08)	(1.57)	(-0.58)

Panel C: Non-transient institutional investors								
	Change in institutional holding				Change in investor number			
	Loser	Middle	Winner	W - L	Loser	Middle	Winner	W - L
t + 1	-0.16%	0.36%	0.19%	0.35%	-0.41	0.54	1.16	1.57
	(-2.44)	(4.41)	(2.42)	(7.05)	(-4.59)	(4.18)	(6.76)	(12.02)
t + 2	-0.17%	0.24%	0.29%	0.46%	-0.27	0.46	1.07	1.34
	(-2.56)	(2.87)	(3.68)	(9.80)	(-2.91)	(3.41)	(6.47)	(11.38)
t + 3	-0.15%	0.17%	0.28%	0.43%	-0.19	0.41	0.99	1.18
	(-2.15)	(1.96)	(3.63)	(9.10)	(-2.08)	(2.93)	(5.94)	(9.81)
t + 4	-0.11%	0.15%	0.22%	0.34%	-0.10	0.40	0.87	0.96
	(-1.59)	(1.76)	(2.70)	(6.21)	(-1.08)	(2.83)	(4.98)	(7.80)

Table 5: Institutional demand in industry 52-week high portfolios

This table reports quarterly changes in total, transient, and non-transient institutional holding and changes in the number of total, transient, and non-transient institutional investors holding the stocks in industry 52-week high portfolios. Total, transient, or non-transient institutional holding of a stock in a quarter is defined as the number of shares held by total, transient, or non-transient institutional investors at the end of that quarter divided by the number of shares outstanding. For each quarter t , we group all stocks into three industry 52-week high portfolios. The industry 52-week high winner (loser) is the equally weighted portfolio of the stocks in the top (bottom) 6 industries ranked by the industry value-weighted ratio of current price to 52-week high. Industry 52-week high middle portfolio is the equally weighted portfolio of stocks that are neither industry 52-week high winners nor losers. For each portfolio, we report quarterly equal-weighted average of change in institutional holding and change in the number of institutions holding the stock for quarters $t+1$ to $t+4$. Panels B & C report the results for transient and non-transient institutions, respectively. The classification of institutions is from Brian Bushee's website. The t -statistics are in parentheses.

Panel A: All institution investors								
	Change in institutional holding				Change in investor number			
	Loser	Middle	Winner	W - L	Loser	Middle	Winner	W - L
$t + 1$	0.14%	0.27%	0.24%	0.10%	0.35	0.88	1.27	0.92
	(1.29)	(2.88)	(2.57)	(2.10)	(1.67)	(4.88)	(5.02)	(5.24)
$t + 2$	0.15%	0.20%	0.19%	0.04%	0.57	0.75	1.08	0.51
	(1.47)	(2.05)	(1.69)	(0.66)	(2.90)	(4.08)	(4.19)	(2.89)
$t + 3$	0.16%	0.18%	0.17%	0.01%	0.68	0.75	1.00	0.32
	(1.55)	(1.73)	(1.62)	(0.13)	(3.46)	(3.85)	(3.76)	(1.91)
$t + 4$	0.12%	0.15%	0.17%	0.05%	0.57	0.70	1.03	0.46
	(1.14)	(1.42)	(1.65)	(0.98)	(2.95)	(3.72)	(4.05)	(2.88)

Panel B: Transient institutional investors								
	Change in institutional holding				Change in investor number			
	Loser	Middle	Winner	W - L	Loser	Middle	Winner	W - L
$t + 1$	-0.00%	0.09%	0.08%	0.08%	0.11	0.28	0.39	0.29
	(-0.09)	(1.89)	(1.82)	(2.18)	(1.05)	(3.14)	(3.31)	(3.20)
$t + 2$	0.03%	0.06%	0.05%	0.02%	0.25	0.24	0.25	0.00
	(0.73)	(1.20)	(1.09)	(0.61)	(2.65)	(2.56)	(1.98)	(0.03)
$t + 3$	0.08%	0.04%	0.02%	-0.05%	0.28	0.23	0.24	-0.04
	(1.67)	(0.89)	(0.54)	(-1.86)	(2.90)	(2.35)	(2.11)	(-0.54)
$t + 4$	0.06%	0.03%	0.02%	-0.03%	0.26	0.22	0.22	-0.03
	(1.15)	(0.65)	(0.53)	(-1.10)	(2.60)	(2.43)	(1.90)	(-0.50)

Panel C: Non-transient institutional investors								
	Change in institutional holding				Change in investor number			
	Loser	Middle	Winner	W - L	Loser	Middle	Winner	W - L
t + 1	0.12%	0.17%	0.16%	0.03%	0.16	0.50	0.77	0.61
	(1.50)	(2.09)	(1.97)	(0.83)	(1.17)	(4.03)	(4.68)	(6.37)
t + 2	0.11%	0.13%	0.12%	0.01%	0.26	0.42	0.66	0.40
	(1.36)	(1.57)	(1.31)	(0.28)	(1.99)	(3.27)	(3.76)	(3.77)
t + 3	0.05%	0.13%	0.14%	0.09%	0.28	0.43	0.64	0.36
	(0.61)	(1.63)	(1.53)	(1.78)	(2.13)	(3.20)	(3.56)	(3.34)
t + 4	0.05%	0.11%	0.14%	0.09%	0.25	0.40	0.67	0.42
	(0.59)	(1.35)	(1.63)	(2.13)	(1.93)	(2.93)	(3.75)	(4.29)

Table 6: Pairwise comparison of the 52-week high and momentum strategies

This table reports the average monthly returns from July 1963 through December 2009 for equally weighted portfolios. Stocks are sorted independently by past 6-month return and by the 52-week high measure. Individual momentum winners (losers) are the 30% of stocks with the highest (lowest) past 6-month return. Industry momentum winners (losers) are the stocks in the 6 industries with the highest (lowest) past 6-month industry return. Individual 52-week high winners (losers) are the 30% stocks with the highest (lowest) ratio of current price to 52-week high. Industry 52-week high winners (losers) are stocks in the top (bottom) 6 industries ranked by the industry value-weighted ratio of current price to 52-week high. All portfolios are held for 6 months. The t-statistics are in parentheses.

Panel A			
Individual Momentum	Industry 52-Week High	Raw return	DGTW return
Winner	Winner	1.68%	0.28%
	Loser	1.15%	-0.04%
	Winner - Loser	0.52%(4.65)	0.32%(3.67)
Middle	Winner	1.36%	0.25%
	Loser	0.96%	-0.05%
	Winner - Loser	0.40%(4.59)	0.30%(4.85)
Loser	Winner	1.37%	0.36%
	Loser	0.63%	-0.20%
	Winner - Loser	0.74%(5.47)	0.56%(5.23)

Panel B			
Industry 52-Week High	Individual Momentum	Raw return	DGTW return
Winner	Winner	1.68%	0.28%
	Loser	1.37%	0.36%
	Winner - Loser	0.31%(1.66)	-0.08%(-1.23)
Middle	Winner	1.44%	0.14%
	Loser	1.05%	0.11%
	Winner - Loser	0.39%(2.30)	0.03%(0.69)
Loser	Winner	1.15%	-0.04%
	Loser	0.63%	-0.20%
	Winner - Loser	0.52%(2.81)	0.16%(5.23)

Panel C			
Industry Momentum	Industry 52-Week High	Raw return	DGTW return
Winner	Winner	1.60%	0.37%
	Loser	1.18%	0.19%
	Winner - Loser	0.41%(2.24)	0.18%(1.65)
Middle	Winner	1.30%	0.18%
	Loser	0.93%	-0.08%
	Winner - Loser	0.37%(3.21)	0.26%(3.38)
Loser	Winner	1.00%	0.17%
	Loser	0.88%	-0.12%
	Winner - Loser	0.11%(0.56)	0.28%(3.02)

Panel D			
Industry 52-Week High	Industry Momentum	Raw return	DGTW return
Winner	Winner	1.60%	0.37%
	Loser	1.00%	0.17%
	Winner - Loser	0.60%(3.15)	0.21%(1.94)
Middle	Winner	1.41%	0.24%
	Loser	1.17%	0.05%
	Winner - Loser	0.25%(2.02)	0.19%(2.22)
Loser	Winner	1.18%	0.19%
	Loser	0.88%	-0.12%
	Winner - Loser	0.30%(1.66)	0.31%(2.64)

Panel E			
Individual 52-Week High	Industry 52-Week High	Raw return	DGTW return
Winner	Winner	1.44%	0.15%
	Loser	1.24%	0.03%
	Winner - Loser	0.21%(2.23)	0.12%(1.83)
Middle	Winner	1.47%	0.33%
	Loser	0.98%	-0.02%
	Winner - Loser	0.48%(5.29)	0.35%(5.18)
Loser	Winner	1.40%	0.37%
	Loser	0.54%	-0.26%
	Winner - Loser	0.86%(5.89)	0.64%(5.15)

Panel F			
Industry 52-Week High	Individual 52-Week High	Raw return	DGTW return
Winner	Winner	1.44%	0.15%
	Loser	1.40%	0.37%
	Winner - Loser	0.04%(0.17)	-0.22%(-2.18)
Middle	Winner	1.37%	0.12%
	Loser	1.02%	0.09%
	Winner - Loser	0.35%(1.62)	0.03%(0.38)
Loser	Winner	1.24%	0.03%
	Loser	0.54%	-0.26%
	Winner - Loser	0.70%(3.06)	0.29%(3.47)

Table 7: Comparison of JT, MG, individual 52-week high, and industry 52-week high strategies

Each month between July 1963 and December 2009, the following cross-sectional regressions are estimated:

$$R_{it} = b_{0jt} + b_{1jt}R_{i,t-1} + b_{2jt}SIZE_{i,t-1} + b_{3jt}JH_{i,t-j} + b_{4jt}JL_{i,t-j} + b_{5jt}MH_{i,t-j} + b_{6jt}ML_{i,t-j} + b_{7jt}GH_{i,t-j} + b_{8jt}GL_{i,t-j} + b_{9jt}IH_{i,t-j} + b_{10jt}IL_{i,t-j} + e_{it}$$

where $R_{i,t}$ and $SIZE_{i,t}$ are the return and the market capitalization of stock i in month t . $IH_{i,t-j}$ ($IL_{i,t-j}$) is the industry 52-week high winner (loser) dummy that takes the value of 1 if the industry 52-week high measure for stock i is ranked in the top (bottom) 30% in month $t-j$, and is zero otherwise. $GH_{i,t-j}$ ($GL_{i,t-j}$) is the individual 52-week high winner (loser) dummy that takes the value of 1 if the individual 52-week high measure for stock i is ranked in the top (bottom) 30% in month $t-j$, and is zero otherwise. The individual 52-week high measure in month $t-j$ is the ratio of price level in month $t-j$ to the maximum price achieved in months $t-j-12$ to $t-j$. The industry 52-week high measure is the value-weighted individual 52-week high measure. $JH_{i,t-j}$ ($JL_{i,t-j}$) equals to one if stock i 's return over the 6-month period ($t-j-6$, $t-j$) is in the top (bottom) 30%, and is zero otherwise; $MH_{i,t-j}$ ($ML_{i,t-j}$) equals to one if stock i 's value-weighted industry return over the 6-month period ($t-j-6$, $t-j$) is in the top (bottom) 30%, and is zero otherwise. This table reports the average of the month-by-month estimates of $\frac{1}{6}\sum_{j=2}^7 b_{3jt}$, $\dots$, $\frac{1}{6}\sum_{j=2}^7 b_{10jt}$. Numbers in parentheses are t -statistics.

	Raw return			DGTW return		
	Whole	Jan. Excl.	Jan. Only	Whole	Jan. Excl.	Jan. Only
Intercept	0.0204 (7.37)	0.0125 (4.87)	0.1075 (9.56)	0.0061 (7.50)	0.0064 (7.64)	0.0031 (0.91)
$R_{i,t-1}$	-0.0560 (-15.47)	-0.0468 (-14.59)	-0.1589 (-7.82)	-0.0636 (-22.11)	-0.0590 (-21.19)	-0.1156 (-8.21)
Size	-0.0018 (-5.28)	-0.0007 (-2.38)	-0.0138 (-9.68)	-0.0009 (-7.31)	-0.0010 (-8.00)	-0.0001 (0.15)
JT winner dummy	0.0018 (2.42)	0.0016 (2.03)	0.0043 (1.61)	0.0000 (0.04)	-0.0006 (-1.80)	0.0074 (4.43)
JT loser dummy	-0.0023 (-4.48)	-0.0029 (-5.90)	0.0044 (1.59)	-0.0013 (-5.34)	-0.0010 (-4.02)	-0.0049 (-4.93)
MG winner dummy	0.0017 (2.55)	0.0015 (2.18)	0.0036 (1.66)	0.0011 (1.97)	0.0010 (1.67)	0.0025 (1.27)
MG loser dummy	0.0001 (0.22)	0.0002 (0.43)	-0.0012 (-0.51)	-0.0004 (-0.90)	-0.0003 (-0.62)	-0.0020 (-0.93)
Individual 52-week high winner dummy	0.0013 (2.17)	0.0023 (3.88)	-0.0096 (-3.18)	0.0003 (0.83)	0.0010 (2.55)	-0.0071 (-3.85)
Individual 52-week high loser dummy	-0.0040 (-3.38)	-0.0071 (-6.53)	0.0306 (5.37)	-0.0022 (-3.43)	-0.0036 (-5.90)	0.0132 (4.51)
Industry 52-week high winner dummy	0.0017 (3.30)	0.0014 (2.77)	0.0045 (1.95)	0.0013 (3.05)	0.0013 (2.87)	0.0022 (1.05)
Industry 52-week high loser dummy	-0.0017 (-3.07)	-0.0018 (-3.17)	-0.0009 (-0.35)	-0.0012 (-2.51)	-0.0013 (-2.50)	-0.0009 (-0.45)
JT winner dummy - JT loser dummy	0.0041 (4.12)	0.0045 (4.37)	-0.0001 (-0.02)	0.0014 (2.54)	0.0004 (0.73)	0.0123 (5.33)
MG winner dummy - MG loser dummy	0.0016 (1.70)	0.0013 (1.33)	0.0048 (1.48)	0.0016 (1.97)	0.0013 (1.57)	0.0045 (1.59)
Individual 52-week high winner dummy - Individual 52-week high loser dummy	0.0054 (3.16)	0.0094 (5.95)	-0.0401 (-5.05)	0.0025 (2.62)	0.0046 (4.97)	-0.0203 (-4.65)
Industry 52-week high winner dummy - Industry 52-week high loser dummy	0.0034 (4.42)	0.0032 (4.07)	0.0054 (1.75)	0.0026 (3.96)	0.0025 (3.77)	0.0031 (1.21)

Table 8: Long-run return of the industry 52-week high strategy

Each month between July 1963 and December 2009, the following cross-sectional regressions are estimated:

$$R_{it} = b_{0jt} + b_{1jt}R_{i,t-1} + b_{2jt}SIZE_{i,t-1} + b_{3jt}JH_{i,t-j-k} + b_{4jt}JL_{i,t-j-k} + b_{5jt}MH_{i,t-j-k} + b_{6jt}ML_{i,t-j-k} + b_{7jt}GH_{i,t-j-k} + b_{8jt}GL_{i,t-j-k} + b_{9jt}IH_{i,t-j-k} + b_{10jt}IL_{i,t-j-k} + e_{it}$$

where $R_{i,t}$ and $SIZE_{i,t}$ are the return and the market capitalization of stock i in month t . $IH_{i,t-j-k}$ ($IL_{i,t-j-k}$) is the industry 52-week high winner (loser) dummy that takes the value of 1 if the industry 52-week high measure for stock i is ranked in the top (bottom) 30% in month $t-j-k$, and is zero otherwise. $GH_{i,t-j-k}$ ($GL_{i,t-j-k}$) is the individual 52-week high winner (loser) dummy that takes the value of 1 if the individual 52-week high measure for stock i is ranked in the top (bottom) 30% in month $t-j-k$, and is zero otherwise. The individual 52-week high measure in month $t-j-k$ is the ratio of price level in month $t-j-k$ to the maximum price achieved in months $t-j-k-12$ to $t-j-k$. The industry 52-week high measure is the value-weighted individual 52-week high measure. $JH_{i,t-j-k}$ ($JL_{i,t-j-k}$) equals to one if stock i 's return over the 6-month period ($t-j-k-6$, $t-j-k$) is in the top (bottom) 30%, and is zero otherwise; $MH_{i,t-j-k}$ ($ML_{i,t-j-k}$) equals to one if stock i 's value-weighted industry return over the 6-month period ($t-j-k-6$, $t-j-k$) is in the top (bottom) 30%, and is zero otherwise. The index k determines the time gap across which persistence is measured. In the table, $k = 12, 24, 36$. Table 5 reports the average of the month-by-month estimates of $\frac{1}{6}\sum_{j=2}^7 b_{3jt}, \dots, \frac{1}{6}\sum_{j=2}^7 b_{10jt}$. Numbers in parentheses are t -statistics.

	$k = 12$		$k = 24$		$k = 36$	
	Raw ret.	DGTW ret.	Raw ret.	DGTW ret.	Raw ret.	DGTW ret.
Intercept	0.0183 (6.23)	0.0038 (4.27)	0.0185 (5.82)	0.0039 (4.59)	0.0181 (5.53)	0.0037 (4.20)
$R_{i,t-1}$	-0.0575 (-14.73)	-0.0642 (-22.42)	-0.0575 (-14.53)	-0.0653 (-23.13)	-0.0584 (-14.56)	-0.0659 (-23.14)
Size	-0.0012 (-3.36)	-0.0005 (-3.42)	-0.0012 (-3.07)	-0.0004 (-3.31)	-0.0011 (-2.88)	-0.0005 (-3.59)
JT winner dummy	-0.0015 (-2.32)	-0.0009 (-2.62)	-0.0011 (-1.91)	-0.0006 (-1.57)	-0.0007 (-1.23)	-0.0005 (-1.26)
JT loser dummy	0.0005 (1.54)	0.0006 (2.17)	0.0005 (1.64)	0.0005 (2.23)	0.0007 (2.28)	0.0008 (2.91)
MG winner dummy	-0.0015 (-2.34)	-0.0017 (-3.23)	-0.0007 (-1.16)	-0.0005 (-1.01)	0.0005 (0.71)	0.0004 (0.81)
MG loser dummy	-0.0001 (-0.20)	-0.0000 (-0.07)	-0.0004 (-0.70)	-0.0007 (-1.31)	0.0000 (0.00)	0.0001 (0.22)
Individual 52-week high winner dummy	-0.0001 (-0.15)	0.0002 (0.48)	-0.0003 (-0.66)	-0.0003 (-0.97)	-0.0003 (-0.60)	-0.0004 (-1.36)
Individual 52-week high loser dummy	0.0010 (0.92)	0.0014 (1.87)	0.0012 (1.22)	0.0015 (2.22)	0.0001 (0.17)	0.0007 (1.27)
Industry 52-week high winner dummy	0.0001 (0.29)	0.0003 (0.61)	-0.0005 (-0.79)	0.0001 (0.17)	-0.0005 (-0.88)	-0.0004 (-0.73)
Industry 52-week high loser dummy	-0.0006 (-1.08)	-0.0006 (-1.32)	-0.0004 (-0.76)	-0.0007 (-1.42)	-0.0006 (-0.96)	-0.0005 (-1.07)
JT winner dummy -	-0.0020 (-2.73)	-0.0015 (-3.22)	-0.0016 (-2.30)	-0.0011 (-2.41)	-0.0014 (-2.11)	-0.0012 (-2.54)
JT loser dummy	-0.0014 (-1.54)	-0.0016 (-2.18)	-0.0003 (-0.36)	0.0001 (0.17)	0.0005 (0.52)	0.0003 (0.43)
MG winner dummy -	-0.0011 (-0.69)	-0.0012 (-1.19)	-0.0015 (-1.09)	-0.0018 (-1.94)	-0.0004 (-0.34)	-0.0012 (-1.47)
MG loser dummy	0.0008 (0.97)	0.0009 (1.39)	-0.0001 (-0.09)	0.0007 (1.11)	0.0000 (0.05)	0.0001 (0.28)
Individual 52-week high winner dummy -						
Individual 52-week high loser dummy						
Industry 52-week high winner dummy -						
Industry 52-week high loser dummy						

Table 9: Profits of the individual 52-week high strategy of firms with different industry betas and R-squares

This table reports the average monthly portfolio returns for individual 52-week high strategy for each tercile which is ranked by the R-square or industry beta ($\beta_{ind,i}$) from the regression $R_{i,t} = a_i + \beta_{mkt,i} R_{m,t} + \beta_{ind,i} R_{ind,t} + e_{i,t}$, where $R_{i,t}$ is the return of stock i on day t , $R_{m,t}$ is the market return on day t , and $R_{ind,t}$ is the value-weighted stock return of stock i 's industry. We run this regression at the end of each month for each stock, using returns in the past year. Each month, stocks are sorted by R-square or industry beta ($\beta_{ind,i}$) from this regression. Individual 52-week high winner (loser) portfolio is the equal-weighted portfolio of the 30% of stocks with the highest (lowest) ratio of current price to 52-week high. The monthly returns are from July 1963 to December 2009. Numbers in parentheses are t -statistics.

Panel A: Rank by industry beta						
	Raw ret.			DGTW-adjusted ret.		
	T1-Low	T2	T3-High	T1-Low	T2	T3-High
Winner	1.39%	1.32%	1.32%	0.13%	0.08%	0.11%
	(7.62)	(7.81)	(6.25)	(3.53)	(2.02)	(3.46)
Loser	1.06%	0.92%	0.81%	0.12%	-0.01%	-0.03%
	(3.09)	(3.07)	(1.97)	(2.01)	(-0.31)	(-0.39)
Winner-Loser	0.32%	0.40%	0.51%	0.01%	0.09%	0.15%
	(1.43)	(2.12)	(1.78)	(0.12)	(1.47)	(1.45)

Panel B: Rank by R-square						
	Raw ret.			DGTW-adjusted ret.		
	T1-Low	T2	T3-High	T1-Low	T2	T3-High
Winner	1.39%	1.37%	1.28%	0.09%	0.12%	0.11%
	(8.85)	(7.18)	(6.03)	(1.30)	(3.40)	(3.69)
Loser	1.44%	0.81%	0.48%	0.35%	-0.05%	-0.21%
	(3.96)	(2.18)	(1.31)	(4.40)	(-0.81)	(-2.57)
Winner-Loser	-0.05%	0.56%	0.80%	-0.26%	0.16%	0.33%
	(-0.19)	(2.28)	(3.37)	(-2.44)	(2.05)	(3.82)

Table 10: Profits of the industry 52-week high strategy for firms with different price informativeness measures

This table reports the average monthly portfolio returns for the industry 52-week high strategy for each group which is ranked by the price informativeness measures: size, age, price impact, analyst coverage, and institutional ownership. Industry 52-week high winners (losers) are stocks in the top (bottom) 6 industries ranked by the industry value-weighted ratio of current price to 52-week high. Each month, within each industry, stocks are sorted into three groups by size, age, price impact, analyst coverage, and institutional ownership. All portfolios are held for 6 months. The t-statistics are in parentheses.

Panel A: Size and industry 52-week High (July 1963 - December 2009)									
Size	Raw return			DGTW return			4-factor return		
	Winner	Loser	W-L	Winner	Loser	W-L	Winner	Loser	W-L
T1 - Small	1.85%	1.11%	0.74%	0.49%	-0.04%	0.53%	0.62%	0.22%	0.41%
	(6.70)	(3.61)	(5.37)	(7.06)	(-0.60)	(5.12)	(4.34)	(1.29)	(3.13)
T2	1.34%	0.68%	0.66%	0.21%	-0.20%	0.40%	0.07%	-0.20%	0.26%
	(5.49)	(2.40)	(4.75)	(4.10)	(-4.15)	(4.96)	(0.88)	(-1.97)	(2.33)
T3 - Large	1.23%	0.82%	0.41%	0.17%	-0.07%	0.24%	0.06%	0.06%	0.00%
	(5.83)	(3.21)	(3.02)	(4.08)	(-1.27)	(3.10)	(1.02)	(0.78)	(0.02)
Panel B: Age and industry 52-week high (July 1963 - December 2009)									
Age	Raw ret			DGTW ret			4-factor return		
	Winner	Loser	W-L	Winner	Loser	W-L	Winner	Loser	W-L
T1 - Small	1.44%	0.69%	0.76%	0.30%	-0.15%	0.46%	0.22%	-0.06%	0.28%
	(5.56)	(2.28)	(4.89)	(4.41)	(-2.04)	(4.49)	(2.09)	(-0.49)	(2.170)
T2	1.55%	0.88%	0.66%	0.30%	-0.14%	0.44%	0.31%	0.02%	0.29%
	(6.37)	(3.11)	(4.69)	(5.59)	(-2.51)	(5.07)	(3.52)	(0.16)	(2.45)
T3 - Large	1.42%	1.02%	0.40%	0.25%	-0.04%	0.29%	0.20%	0.10%	0.10%
	(6.68)	(4.17)	(3.53)	(5.52)	(-0.96)	(4.17)	(3.07)	(1.33)	(1.13)
Panel C: Price impact and 52-week high (July 1963 - December 2009)									
Price impact	Raw ret			DGTW ret			4-factor return		
	Winner	Loser	W-L	Winner	Loser	W-L	Winner	Loser	W-L
T1 - Small	1.15%	0.74%	0.41%	0.16%	-0.09%	0.25%	0.00%	0.03%	-0.03%
	(5.37)	(2.74)	(2.76)	(3.62)	(-1.41)	(3.04)	(0.00)	(0.36)	(-0.28)
T2	1.33%	0.76%	0.57%	0.20%	-0.18%	0.38%	0.08%	-0.10%	0.18%
	(5.46)	(2.62)	(4.01)	(3.82)	(-3.64)	(4.70)	(1.01)	(-0.98)	(1.56)
T3 - Large	1.87%	1.20%	0.67%	0.49%	0.04%	0.45%	0.56%	0.17%	0.39%
	(6.84)	(4.02)	(5.33)	(7.05)	(0.56)	(4.58)	(4.17)	(1.07)	(3.34)

Panel D: Analyst coverage and 52-week high (January 1984 - December 2009)									
Analyst	Raw ret			DGTW ret			4-factor return		
	Winner	Loser	W-L	Winner	Loser	W-L	Winner	Loser	W-L
No	1.40%	0.76%	0.65%	0.42%	-0.07%	0.49%	0.39%	0.15%	0.23%
	(4.25)	(1.98)	(2.77)	(4.10)	(-0.79)	(3.42)	(2.15)	(0.81)	(1.21)
Small	1.49%	0.78%	0.72%	0.35%	-0.17%	0.51%	0.40%	0.04%	0.36%
	(4.63)	(2.01)	(3.42)	(4.47)	(-2.33)	(4.25)	(3.09)	(0.29)	(2.16)
Large	1.34%	0.95%	0.39%	0.26%	-0.01%	0.27%	0.18%	0.20%	-0.02%
	(4.67)	(2.43)	(1.59)	(3.57)	(-0.08)	(1.91)	(1.91)	(1.78)	(-0.14)

Panel E: Institutional ownership and 52-week high (January 1980 - December 2009)									
IO	Raw ret			DGTW ret			4-factor return		
	Winner	Loser	W-L	Winner	Loser	W-L	Winner	Loser	W-L
T1 - Small	1.50%	0.73%	0.77%	0.51%	-0.12%	0.63%	0.39%	0.07%	0.32%
	(4.87)	(1.99)	(3.51)	(5.09)	(-1.34)	(4.82)	(2.07)	(0.32)	(1.64)
T2	1.54%	0.72%	0.82%	0.31%	-0.21%	0.52%	0.31%	-0.02%	0.33%
	(5.38)	(2.04)	(4.05)	(4.44)	(-2.78)	(4.43)	(2.60)	(-0.11)	(2.09)
T3 - Large	1.42%	0.89%	0.54%	0.19%	-0.10%	0.29%	0.13%	0.00%	0.13%
	(5.33)	(2.71)	(2.95)	(2.96)	(-1.27)	(2.52)	(1.46)	(-0.03)	(0.96)

Table 11: Portfolio rebalancing and individual and industry 52-week high strategies

Panel A reports returns to individual and industry 52-week high strategies if we do not rebalance the portfolio. Each month, we form portfolios based on individual and industry 52-week high measures and hold the portfolios for six months without rebalancing. Then we calculate the buy and hold six-month cumulative raw return and the buy and hold six-month cumulative abnormal return, where the abnormal return is the six-month cumulative raw return minus the six month cumulative raw return on the size/book-to-market ratio/momentum portfolio. Panel B reports monthly value-weighted average portfolio returns for small stocks. Each month, we form portfolios based on the 52-week high measures and then calculate monthly value-weighted average small stock returns for each portfolio. Small stocks are stocks with size below 25 percentile of all stocks. All portfolios are held for 6 months. The sample includes all stocks on CRSP from July 1963 through December 2009; t-statistics are in parentheses.

Panel A: Monthly returns without rebalancing						
	Raw return			DGTW return		
	Winner	Loser	Winner-Loser	Winner	Loser	Winner-Loser
Individual	1.27%	0.43%	0.84%	0.08%	-0.00%	0.09%
	(14.21)	(2.62)	(8.62)	(5.80)	(-0.18)	(2.56)
Industry	1.28%	0.62%	0.66%	0.25%	-0.14%	0.39%
	(10.70)	(4.93)	(11.80)	(8.98)	(-4.81)	(8.60)

Panel B: Monthly value-weighted average portfolio return among small stocks (Size <= 25 percentile)						
	Raw return			DGTW return		
	Winner	Loser	Winner-Loser	Winner	Loser	Winner-Loser
Individual	1.58%	0.69%	0.89%	0.06%	-0.27%	0.33%
	(8.17)	(1.93)	(3.97)	(1.46)	(-6.35)	(4.58)
Industry	1.56%	0.63%	0.93%	0.16%	-0.36%	0.52%
	(5.34)	(2.16)	(6.56)	(2.55)	(-7.52)	(5.95)

Table 12: Individual and industry 52-week high strategies in different time periods

This table reports the average monthly portfolio returns for individual and industry 52-week high strategies in four time periods. All portfolios are held for 6 months. The winner (loser) portfolio in individual 52-week high strategy is the equally weighted portfolio of the 30% stocks with the highest (lowest) ratio of current price to 52-week high. The winner (loser) portfolio in industry 52-week high strategy is the equally weighted portfolio of the stocks in the top (bottom) 6 industries ranked by the industry value-weighted ratio of current price to 52-week high. The sample includes all stocks on CRSP; *t*-statistics are in parentheses.

		Raw return			DGTW return		
		Winner	Loser	Winner-Loser	Winner	Loser	Winner-Loser
July 63 - Dec 78	Individual	1.16%	1.09%	0.08%	0.08%	-0.06%	0.13%
		(3.29)	(1.77)	(0.22)	(1.92)	(-0.91)	(1.40)
	Industry	1.37%	0.98%	0.38%	0.19%	-0.01%	0.20%
		(3.16)	(2.04)	(3.21)	(4.93)	(-0.29)	(2.70)
Jan 79 - Dec 94	Individual	1.65%	0.78%	0.87%	0.14%	0.06%	0.09%
		(5.32)	(1.62)	(3.16)	(3.58)	(0.89)	(1.03)
	Industry	1.50%	0.82%	0.68%	0.20%	-0.15%	0.35%
		(4.05)	(2.16)	(4.67)	(4.38)	(-2.37)	(3.97)
Jan 95 - Dec 09	Individual	1.22%	0.89%	0.34%	0.11%	0.10%	0.01%
		(4.35)	(1.19)	(0.58)	(1.78)	(0.79)	(0.06)
	Industry	1.54%	0.80%	0.75%	0.45%	-0.15%	0.60%
		(3.71)	(1.46)	(2.18)	(3.80)	(-1.38)	(3.16)
Exclude 98 99 00 08 09	Individual	1.46%	0.99%	0.48%	0.11%	0.01%	0.10%
		(7.77)	(2.76)	(2.00)	(4.31)	(0.29)	(1.53)
	Industry	1.50%	0.99%	0.51%	0.22%	-0.07%	0.28%
		(6.55)	(3.58)	(4.13)	(6.29)	(-1.68)	(4.41)

Table 13: The industry 52-week high strategy with alternative holding periods

This table reports the average monthly portfolio returns for individual and the industry 52-week high strategies. The portfolios are held for 3 months (Panel A) or 12 months (Panel B). The winner (loser) portfolio in individual 52-week high strategy is the equally weighted portfolio of the 30% stocks with the highest (lowest) ratio of current price to 52-week high. The winner (loser) portfolio in industry 52-week high strategy is the equally weighted portfolio of the stocks in the top (bottom) 6 industries ranked by the industry value-weighted ratio of current price to 52-week high. The sample includes all stocks on CRSP; *t*-statistics are in parentheses.

Panel A: Hold the portfolio for 3 months							
		Raw return			DGTW return		
		Winner	Loser	Winner-Loser	Winner	Loser	Winner-Loser
whole	Individual	1.35% (7.45)	0.91% (2.52)	0.44% (1.78)	0.09% (3.08)	0.04% (0.89)	0.05% (0.63)
	Industry	1.54% (6.55)	0.76% (2.78)	0.78% (5.74)	0.34% (6.97)	-0.18% (-3.79)	0.53% (6.48)
Jan excluded	Individual	1.22% (6.54)	0.03% (0.09)	1.19% (5.60)	0.14% (5.23)	-0.08% (-1.90)	0.23% (3.45)
	Industry	1.19% (5.02)	0.25% (0.96)	0.93% (6.85)	0.33% (6.62)	-0.23% (-4.57)	0.56% (6.60)
Jan only	Individual	2.80% (4.02)	10.68% (6.02)	-7.87% (-5.67)	-0.53% (-3.71)	1.47% (5.91)	-1.99% (-5.55)
	Industry	5.51% (5.79)	6.37% (5.06)	-0.86% (-1.36)	0.43% (2.16)	0.30% (1.60)	0.13% (0.50)

Panel B: Hold the portfolio for 12 months							
		Raw return			DGTW return		
		Winner	Loser	Winner-Loser	Winner	Loser	Winner-Loser
whole	Individual	1.29% (6.99)	1.04% (3.01)	0.25% (1.16)	0.09% (3.33)	0.10% (1.87)	-0.01% (-0.13)
	Industry	1.41% (6.06)	0.97% (3.62)	0.44% (4.07)	0.23% (5.75)	-0.04% (-0.97)	0.26% (4.06)
Jan excluded	Individual	1.13% (5.99)	0.19% (0.60)	0.94% (4.99)	0.14% (5.32)	-0.04% (-0.92)	0.18% (2.64)
	Industry	1.04% (4.48)	0.47% (1.80)	0.57% (5.15)	0.21% (5.21)	-0.08% (-1.83)	0.28% (4.13)
Jan only	Individual	3.11% (4.17)	10.48% (6.35)	-7.37% (-6.42)	-0.47% (-4.29)	1.67% (6.91)	-2.14% (-6.60)
	Industry	5.55% (5.74)	6.51% (5.37)	-0.97% (-2.24)	0.45% (2.46)	0.36% (2.87)	0.09% (0.39)